

Re-evaluating Swiss reference dates and studying phenological shifts using four independent regional monitoring datasets

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1 Introduction

Data collected from national monitoring schemes play an important role in exploring forces driving changes in population trends of relatively abundant and widespread avian species in Europe. Having an accurate assessment of the two subsets of a bird population (breeding and non-breeding) aids the formulation of conservation policies. However, it depends on the temporal window being chosen to conduct the survey within. In Central European monitoring schemes, a *Stichdatum* (reference date, hereafter RD) tries to minimise the influence of migrants on the counts of the surveys. Essentially, for a species, it is the date, after which the proportion of breeders dominate that of the non-breeders. Population trends and breeding territories are obtained predominantly from surveys, which are conducted, using the species-specific RDs.

RDs in Switzerland had been defined in the 1990s using field knowledge and the limited surveying data available at that point in time. There are two lines of evidence pointing towards the fact that the RDs might be outdated for some species. Firstly, studies have documented changes in phenology in lower trophic levels due to climate change, consequently resulting in a shift in the spring phenology of birds. The changes in phenology

cannot be generalised across birds, as the responses of short-distance and long-distance migrants were not in agreement in all scenarios. Secondly, multiple bird surveying programs across Central Europe have now accumulated data worth multiple years and decades. The resultant time series is bound to provide greater empirical resolution of phenological dates with better precision in comparison to the time when the initial RDs were laid out.

There is an absence of a systemic evaluation of RDs using data from surveys conducted by the Swiss, German or the wider Central European monitoring framework. It is also imperative to ask if such an evaluation would be reproducible if repeated in a neighbouring region, using an independently collected dataset. If breeders are reaching the peak of their activity before the RD, the surveying effort would be missing out on the period truly representative of the breeding population. Conversely, if migrants are departing later, an outdated RD would be capturing the passing tail of the migrant wave, biasing the population estimates upwards. Moreover, even if a robust discrepancy is seen between the RDs and observed phenology, a similar discordance of the same sign and magnitude, might not be recovered from another independent dataset.

Here, we try to re-evaluate the RDs using four independent regional monitoring datasets spanning bavaria, Baden-Württemberg survey programs and Switzerland. Although they differ in spatial and temporal coverages alongside the classification criterion of breeding and non-breeding sites, they do share a common structure. Throughout the spring season, repeated visits to specific survey sites allow us to aggregate an abundance count of different species and provide enough information to differentiate between breeding and non-breeding sites. This was used to address dataset and region-specific questions and introduce a cross dataset analysis.

2 Objectives

Q1. How do empirically estimated breederpeak dates compare with the established reference dates?

Q2. Do the reference dates correspond to the departure of passage migrants, as intended?

Q3. Have breeder and migrant peak dates shifted over the years?

A corresponding subject, which needs attention after these are: are the responses to the aforementioned questions consistent across the datasets?

By addressing these questions at within-species, across-species levels and comparing them across datasets, the goal is to provide a basis for updating the RDs and to incorporate the phenological trends and robustness of information into the future revisions of survey protocols.

3 Materials and Methods

3.1 Overview of datasets

Table 1 provides a summary of the datasets' composition. The reference DOYs used across datasets are in tandem with the Swiss ones. It also contains the empirically estimated phenological dates (breeder peak, migrant peak and the day migrant abundance falls to 50% of its peak value). All datasets record repeated site visits within spring and provide a species-level indicator to separate the population into breeding and non-breeding sites. These metrics of differentiation vary across the independent datasets.

Baden-Württemberg I archive spans 2006 to 2023 but every record from 2006 to 2017 lacks a valid survey date and only contains the year of survey. Thus, the usable analytical window is 2018 to 2023, which happens to be considerably shorter than the Swiss dataset used to analyse the phenological trend across years.

3.2 Shared Conceptual Framework

The fundamental challenge lies in separating migrants from breeders in the monitoring data as they co-occur at survey sites during the spring passage period. While indicators of

Table 1: Overview of the five component analyses.

Dataset	Region	Years	Sites	Species	Classification
Bavaria (MhB-BY)	Bavaria, DE	2020–2024	244	116	Territory count > 0
Swiss migration	Switzerland	2021–2025	1 km ² grid	82	Atlas code ≥ 4 and < 20 or $= 50$; 5 km buffer
Swiss phenology trend	Switzerland	2007–2025 (15 windows)	1 km ² grid	80–83	Atlas code ≥ 4 and < 20 or $= 50$; 5 km buffer
Baden-Württemberg I	Baden-Württemberg, DE	2018–2023	303	130	Behavioural codes, pooled across years
Baden-Württemberg II	Baden-Württemberg, DE (Ornitho)	2020–2025	342	134	Atlas code B/C prefix, pooled across years

breeding activity vary across datasets, the foundational basis of the classification applied here remains the same. A site where reproductive behaviour was strongly indicated for a species, becomes a breeding site for that year. All individuals observed in that site for the time period, are classified under breeding sites. A combination of site $\times$ year $\times$ species has been used for the purpose of designating breeder sites. A non-breeding site for a species $\times$ year combination is one, where no breeding activity has been denoted for the entirety of that year. Phenological curves are fitted independently to the two sites' aggregated daily (or pentade-level) abundance. The peak of the non-breeding curve estimates the date (DOY: Day of Year: 32 being the 1st of February) of maximum migrant abundance; its descendance to 50% of that peak gives the date by which majority of the migrants have departed, while the peak of the breeding curves gives an idea of the date of maximum breeder activity. All these obtained dates are compared with species-specific RDs, denoted as *Reference DOY* – [*observed DOY*]. A negative value means that the RD precedes the observed event and a positive value indicates the RD follows it. Two things ought to be added here: firstly, the counts are expected to be highest just before breeding starts co-inciding with the peak of courtship behaviour (thus, the peak of abundance is a good indicator for the initiation of the breeding season); secondly, migrant individuals can also be observed during the monitoring (thus, an assumption is made that there is an

early migrant abundance peak).

3.3 Bavarian dataset: Territory-based site classification

Data

The dataset records bird surveys at 244 fixed monitoring sites, 2020-2024 (78,290 records). Each visit records the date, species (EURING code and scientific name), number of individuals seen and the number of confirmed territories. RDs for 111 species were available as day-month strings (eg: 15.04.) and converted to DOYS.

Data preparation

Raw survey records were structured into a site $\times$ date $\times$ species matrix to aid the classification into breeding and non-breeding sites. As the dataset only recorded presences via observations, absences had to be incorporated into the dataset. For a species $\times$ site $\times$ year combination, if any count was > 0 , for every other date, the count is set to 0, if the species had not been observed.

Territory-based breeding/non-breeding separation

A site $\times$ year combination was classified as a **breeding site** if the maximum number of territories recorded for that year is > 0 , and as a **non-breeding site** otherwise. This classification is not pooled across years. A site may be breeding and non-breeding in different years.

Phenological curve fitting

LOESS smoothing (span = 0.75) was applied independently to daily abundance from breeding sites and non-breeding sites over DOY 60-190. In the predictive model, certain DOYs were predicted to have **number of individuals** as a negative entry. To preserve ecological sense, each of these entries were replaced by 0. Also, since the datasets were

passed through a model and the plots of non-breeding sites and all sites were imposed on the same plot, it was ensured that the non-breeding abundance did not cross the abundance of the species at all sites at the same point of time.

3.4 Swiss dataset: migration timing (2021-2025)

Data and temporal resolution

The Swiss dataset was obtained from `ornitho.ch`, which relied on casual information collected and reported by citizen scientists. Each observation has an assigned breeding evidence code, called the atlas code. The temporal unit is the **pentade**, which is a 5-day period: pentade p corresponds to DOY $5p - 4$ through $5p$, with representative DOY $= 5p - 2$.

OPM/SOPM and site classification

The observed peak maximum (OPM) for every pentade $\times$ species $\times$ grid $\times$ year is the maximum count recorded across all observation events in that year. These OPM values are summed across grid squares to produce the sum of OPM, SOPM, which can serve as an abundance index. Grid squares with a maximum atlas-code ≥ 4 and ≤ 19 or $= 50$, are classified as **breeding sites**, while sites with the other atlas codes are denoted as **non-breeding sites**. Only sites laying outside a 5 km radius of any pre-denoted breeding site were considered to formulate the non-breeding SOPM, to reduce double-counting of breeders.

Curve fitting and date extraction

LOESS (span = 0.4) was fitted to the daily SOPM series. `migrant50_doy`, `nonbreeding_peak_DOY` and `breeder_peak_doy` were extracted per species. The entry was left as NA, if the data could not be extracted for any of these variable from the provided time-frame. The pentade was used as a representative of the DOY and the aforementioned $5p - 2$ was

used. Species-specific reference DOYs (`bb_doy.csv`) match the Bavaria-sourced values used throughout the rest of this study (Section 3.8).

3.5 Swiss dataset: phenological trends (2007-2025)

In order to detect long term changes, the Swiss dataset was re-analysed through seven windows (2007-2009, 2010-2012, ..., 2022-2024, 2025). Within each window, the previous atlas code boundaries were used for classification of sites and the 5km `buffer zones` persisted as well. LOESS (span = 0.2) was used with a Monte Carlo pentade-jitter procedure to ensure that a pentade does not being representative of a single DOY. Since a shift of any metric by 4 days over a decade is ecologically significant, pentades (5-day windows) were not the most reliable resolution for the process. Simulations were run 250 times for each species×year combination. `migrant50_doy`, `nonbreeding_peak_DOY` and `breeder_peak_doy` were extracted for each time-window (`[species]_phenology_shifts.csv`). A linear model,

$$\text{Peak DOY}_w = \alpha + \beta \cdot \bar{y}_w + \varepsilon_w, \quad (1)$$

was fitted per species and per metric (breeder, migrant), where $\bar{y}_w$ is the window's central year (start +2) and β is the trend slope in days yr^{-1} , a negative beta indicates advancement.

3.6 Baden-Württemberg I: Behavioural-code site classification (2018-2023)

Data

The initial Baden-Württemberg dataset has 303 distinct sites and 130 species, spanning 2018 to 2023. Although the data contains records since 2007, the field of `survey date` doesn't contain entries until 2018.

Data preparation

On the basis of species×year×site combinations, we denoted **breeding** and **non-breeding** sites. Depending on the columns denoting behaviours, if any amongst singing, warning, aggressive interaction, nesting material, feeding, nest, juvenile, territory were denoted as true, the combination serves as a breeding site and otherwise, as a non-breeding site. Using the same metric, if a species ever had a count > 0 for a year×site combination, if it was not observed during other visits that year, it's count is artificially set to 0- but only on dates a real survey visit is known to have occurred at that site (never on unvisited calendar days). A visit-based LOESS (span = 0.75) was fitted directly to these raw, zero-filled site×year×DOY counts over DOY 60-181, with no averaging across years or summing across sites beforehand, every real visit enters the fit as its own observation. From each curve, **breeder peak** and **migrant peak** were obtained.

3.7 Baden-Württemberg II: Atlas code breeder classification (2020-2025)

The second dataset from Baden-Württemberg comprises 342 distinct sites, 134 species and covers a time frame of 2020-2025. Similar to the Swiss dataset, atlas codes were used for classifying into **breeding** and **non-breeding** sites, for species×year×site combinations. It is done on the basis of a confirmed/probable breeding code. Using the same visit-based LOESS approach as the first Württemberg dataset (raw, zero-filled counts, DOY 60–181, no cross-year/cross-site averaging), the abundance plots show how the real collected data maps on to the predicted data, alongside producing the DOYs for breeder and migrant peaks.

3.8 Cross-dataset validation

In order to validate the results produced by a pipeline on one dataset, it was imperative to investigate if other independent surveys agreed with the latter's results. It is

imperative to note that results were only taken into consideration if they fell within the range of DOY 59 to DOY 181 (the period of interest). For this purpose, four metrics were used: (*ref_minus_breed* = Reference DOY – breeder peak DOY [a negative value means that the breeding population peak has arrived before the reference date and vice versa], (*ref_minus_mig* = Reference DOY – migrant peak DOY [refer to the previous inference], breeder peak DOY, migrant peak DOY. Another classification that we decided to use was that of short distance migrants (*Kst*) and long distance migrants (*Lst*). The visualisations mainly used have been dotplots and scatter plots.

Table 2: Phenological metrics extracted from the LOESS curves.

Metric	Ecological indication
Breeder_Peak_DOY	DOY at which the breeding_sites curve reaches its maxima
Migrant_Peak_DOY	DOY at which the non_breeding_sites curve reaches its maxima
Reference_DOY – Breeder_peak_doy	negative response means that the breeding population peak has arrived after the reference date and vice versa
Reference_DOY – Migrant_peak_doy	positive response indicates that the non-breeding population peak before the reference date and vice versa

1. **Lin’s concordance correlation coefficient (CCC)** uses pearson’s R value and a correction factor penalising for shifts across datasets regardless of similar magnitudes of differences,

$$CCC = \frac{2 \sigma_{xy}}{\sigma_x^2 + \sigma_y^2 + (\mu_x - \mu_y)^2}, \quad (2)$$

where μ , σ^2 , and σ_{xy} are the (population) means, variances, and covariance of the two datasets’ offset values for the shared species; a nonparametric bootstrap (2000 resamples) supplied the 95% confidence interval.

A final version of these analyses were also carried out on 16 species which were common to all the four datasets, to overlook any distortion which could be created by absence of records for species present in only a few surveys. The pipeline and results for every dataset is present in the supplementary information.

4 Results

4.1 Two case studies spanning the divide in migratory strategies and an overview of cross-dataset results

To make sense of the cross-dataset results which follow, we wish to initially present two species representative of the long distance and short distance migrants. The choices were deliberately made from the species which spanned all the datasets, *Jynx torquilla* (Lst) and *Picus canus* (Kst). It is imperative to clarify that they are not representative of the median species but they are the clearest possible illustration of what cross-dataset agreement looks like. However, the internal structure of their results (near identical peak dates independently recovered from the four surveys, albeit with different rules of classification) is the pattern expected to recur, even weakly, across the full species pool in Section 4.7. In the following sections, we show some of the combined results which try to investigate the agreement among datasets for the different metrics.

4.2 *Picus canus* (Kst, reference DOY = 51, 20th February)

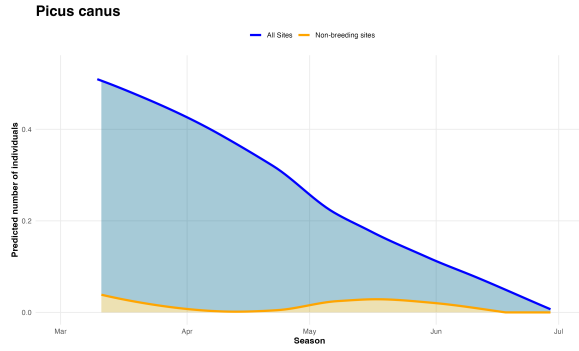
Table 3 summarises the estimated dates recovered from the datasets. Figure 1 shows the visualisations of what we broadly tried to implement across the datasets. Breeder peak estimates range from 70 to 88 and migrant peak dates range from 69 to 90. When the other two columns are looked at, a conclusion can be drawn that the breeding peak and migrants' peak are consistently arriving after the set RD. The independent analyses seem to agree that both the sub-populations seem to reach their maxima well after the RD. The residual migrants' presence is still building up to its maxima way past the RD, which succeeds in capturing a true breeding peak but fails to differentiate the latter from migrant population.

Figure 1(e) shows the distribution of the **breeding** and **non-breeding** sites on a spatial scale across Switzerland. The blue dots signify the spaces where breeding behaviour was indicated, while the red zones indicate the buffer areas to prevent re-counting of breeding

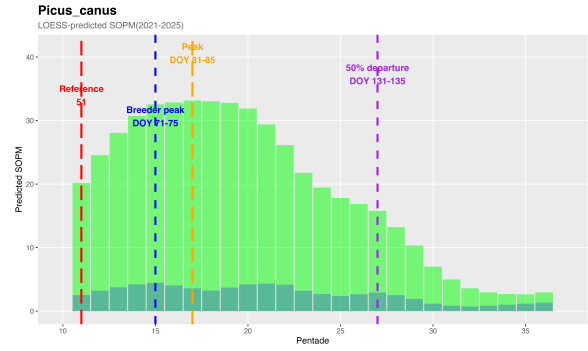
individuals, while the green sites are those where mating behaviour was not reported. Figure 1(f) shows how the population peaks are shifting across the years: breeder peaks are moving forward significantly (population peaks significantly being reached later in comparison to the past) while migrant peaks are moving forward slightly. Figure 1(d) gives an insight to understand how the prediction models are working. Every dot is a raw count which was ever reported for *Picus canus* by the monitoring effort. Depending on the abundances of points on specific DOYs, a predictive figure is plotted about how the population is changing within the target range. The final plot is an artefact supported by raw counts from the dataset.

Table 3: *Picus canus*: empirically estimated peak DOYs per dataset (reference DOY = 51 in all four).

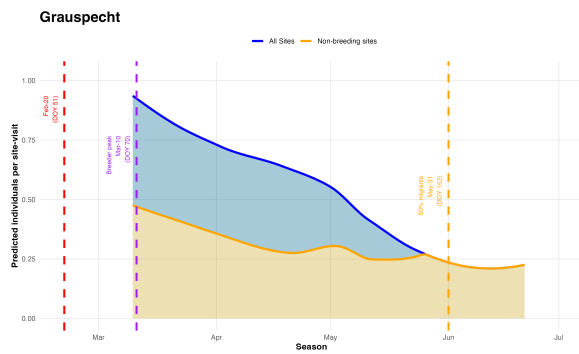
Dataset	Breeder peak DOY	Migrant peak DOY	Ref – breeder peak	Ref – migrant peak
Bavaria	81	70	−30	−19
Swiss migration	73	83	−22	−32
Baden-Württemberg I	70	69	−19	−18
Baden-Württemberg II	88	90	−37	−39



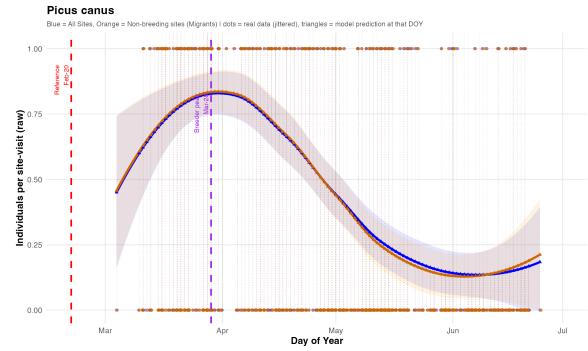
(a) Bavarian dataset: Distribution of the entire population and non-breeders (2020-2024)



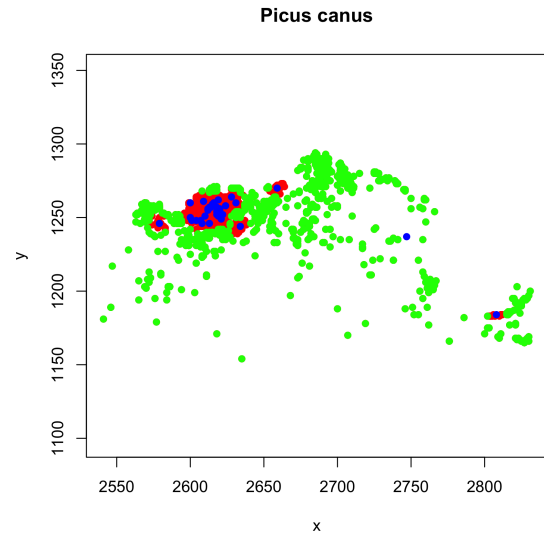
(b) Swiss migration (SOPM, 2021-2025): Pentade-wise distribution



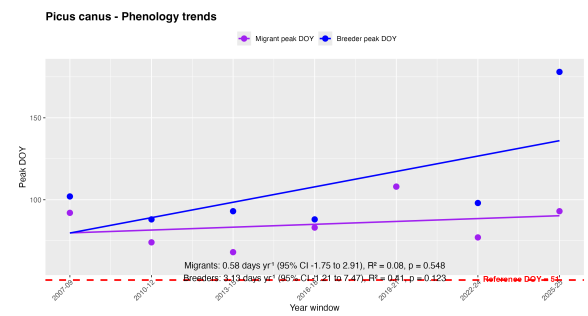
(c) Baden-Württemberg I



(d) Baden-Württemberg II: how the raw abundance counts of the populations map onto the predicted model



(e) Swiss map distribution of breeding and non-breeding sites



(f) Swiss phenology trend, 2007-2025

Figure 1: *Picus canus*: phenology curves across the four surveys, alongside the 2007-25 Swiss trend.

4.2.1 *Jynx torquilla* (Lst, reference DOY = 122, 2nd May)

Jynx torquilla shows an even tighter cross-dataset agreement than *Picus canus*. (Table 4) shows how breeder peak estimates have an overall span of 12 days while that of the migrant peak is 9 days, across the four independently surveyed efforts. Consequently *RD - Breeder peak* and *RD - Migrant peak* show significant agreement.

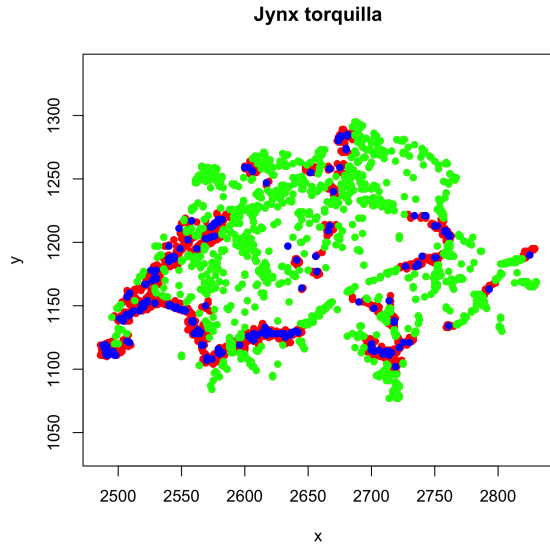
Table 4: *Jynx torquilla*: empirically estimated peak DOYs per dataset (reference DOY = 122 in all four).

Dataset	Breeder peak DOY	Migrant peak DOY	Ref – breeder peak	Ref – migrant peak
Bavaria	122	109	0	13
Swiss migration	113	113	–9	–9
Baden-Württemberg I	110	117	–12	–5
Baden-Württemberg II	113	118	–9	–4

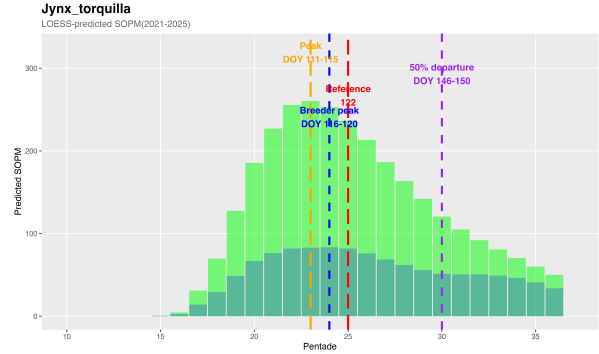
Figure 2(a) shows the distribution of **breeding** and **non-breeding** sites: blue representing the presence of mating individuals, red showing the buffered out zones to avoid recounting of the former while the green sites are those where breeding behaviour was never reported. Figure 2(c) shows how the arrival of the peak dates have been changing over the study period of 2007-2025, and they have stayed more or less constant. Figure 2(d) shows the scaled distribution of the migrants amongst all the Bavarian sites, allowing a statistical representation of how the subpopulations are changing over the time period. Figure 2(f) gives a representation of how the raw abundance density feed themselves into the model resulting in the predicted plot.

4.3 An overview of cross-dataset agreement analysis

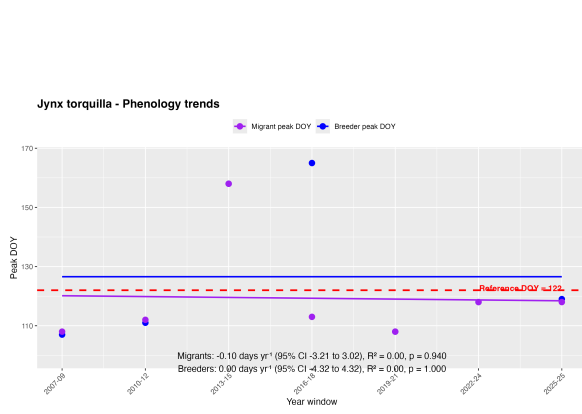
Figure 3 shows segments from the collective analyses that were looked into. Within the Swiss dataset, the phenological shift of migrant peak DOYs behaves differently for long and short distance migrants, as shown in figure 3(d). The forward movement of migrant peak is more pronounced for the short distance migrants. Here, a median slope of -0.57 indicates that over a decade, the arrival of the migrant peak is late by 5.7 days. In figure 3(a) and 3(b), for two different metrics [*RD – Migrant peak DOY*, *RD – Breeder peak DOY*] have been taken into consideration for both migrants to showcase the variety



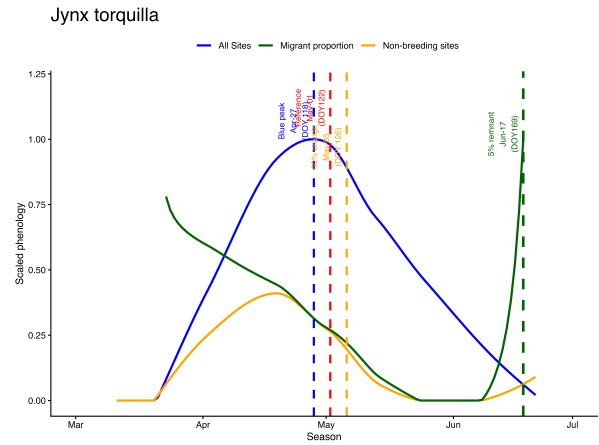
(a) Distribution of breeding, non-breeding and buffer sites imposed on the Swiss map



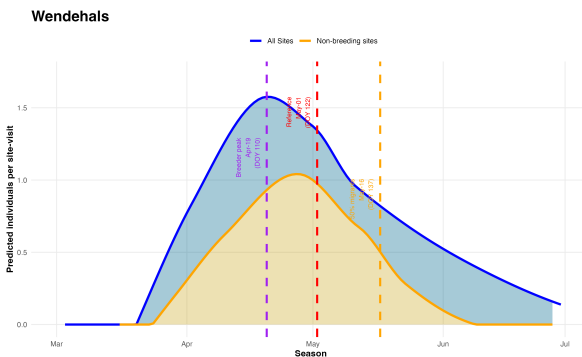
(b) Swiss non-breeding site phenology prediction



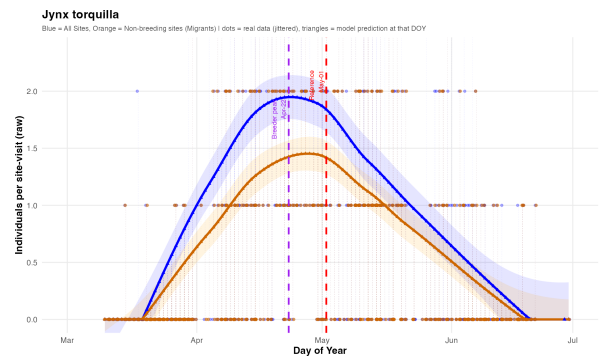
(c) Comparing the abundance of the whole population and the non-breeding subpopulation using the Swiss dataset (2020-2025)



(d) Scaled phenology showing how the migrant proportion in the population changes, according to the Bavarian dataset

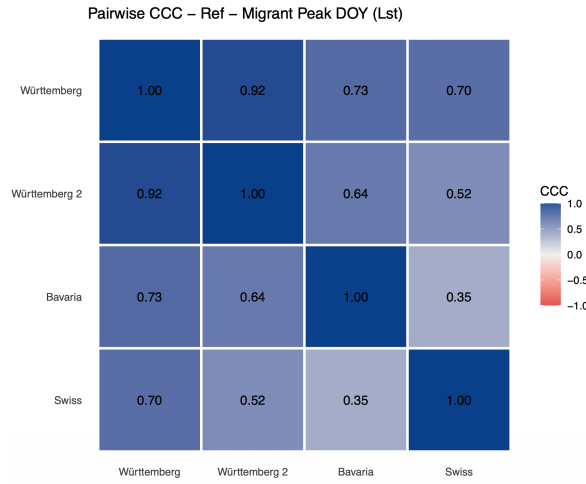


(e) Württemberg 1: all individuals vs migrants

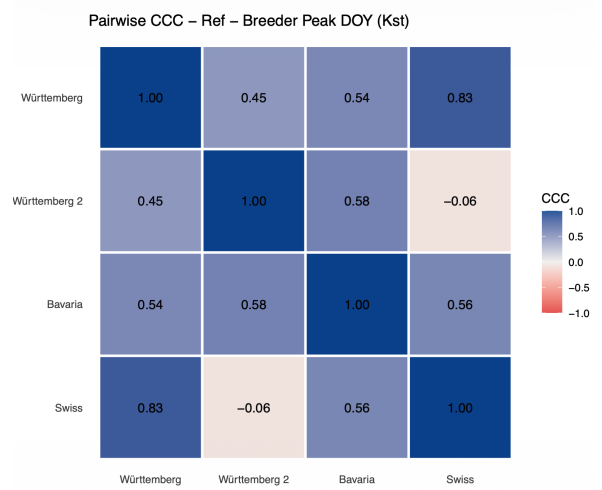


(f) Württemberg 2: visualising how the raw counts translate into the predicted model

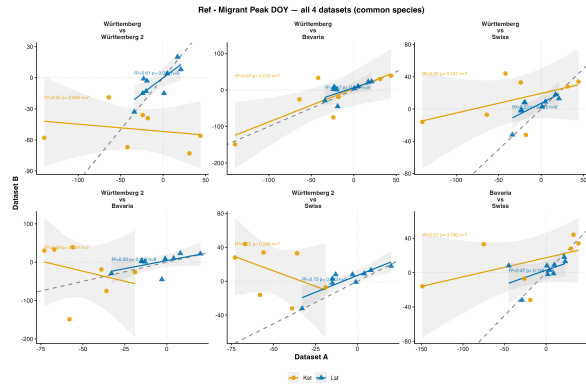
Figure 2: Phenology results for *Jynx torquilla* across the four datasets (repeated view).



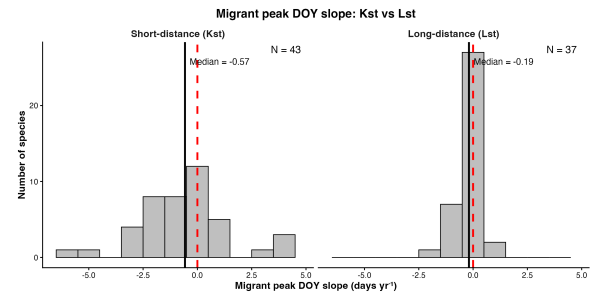
(a) Heatmap of agreement across datasets for the metric, RD – Migrant peak DOY for long distance migrants.



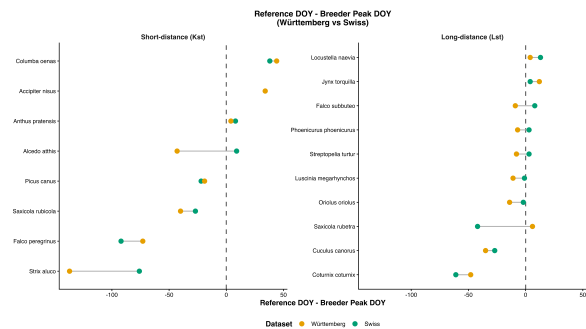
(b) Heatmap of agreement across datasets for the metric, RD – Breeder peak DOY for short distance migrants.



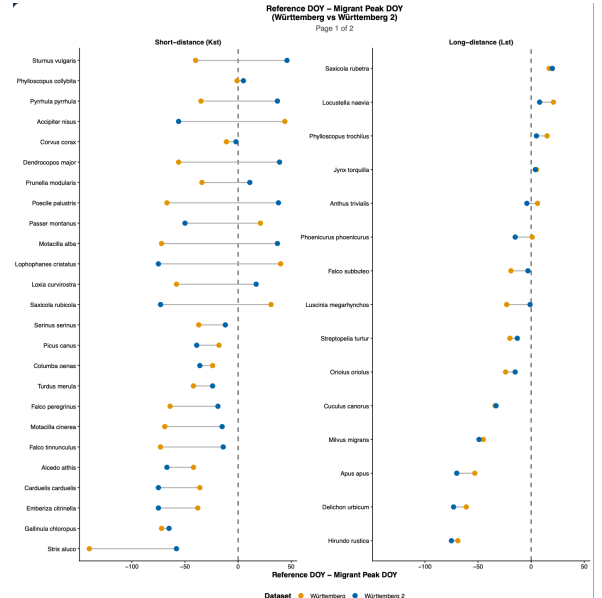
(c) Scatter plot visualising the agreement across datasets for the metric, RD – Migrant peak DOY, across the possible combinations of datasets, for both short and long distance migrants.



(d) Histogram showing how arrival of the migrant peak has shifted since 2007, using the Swiss dataset.



(e) Dotplot visualising RD – breeder peak DOY: Baden-Württemberg I and Swiss datasets, which showed the highest agreement for the same metric for short distance migrants



(f) Dotplot visualising RD – migrant peak DOY: Baden-Württemberg I and II datasets, which showed the highest agreement for the same metric for long distance migrants

Figure 3: Cross-dataset comparisons

in agreement within pairs of datasets. For the corresponding best pairs for each metric, dotplots are present in figure 3(e) and 3(f). However, for a fair comparison, where we take an equal look at all pairs, 3(c) uses the metric RD – Migrant peak DOY to showcase the agreements or disagreements, albeit a good fit or not.

4.4 RD versus breeder phenology

Across all datasets, we can see a negative sign of consistency for RD – Breeder peak DOY for the long distance migrants in Table 5, while there is not such a strong trend for the short distance migrants. For lst, the breeder activity is peaking after the RD, allowing surveys to capture a true peak, but the same is not applying to kst, where the effort might easily miss out on a possible mating peak. A negative offset of high magnitude signifies a good attempt at capturing the peak. However, the opposite is seen for the two Württemberg datasets ranging between 40 and 46% negative values. However, the kst and lst specific regional findings open a can of worms, which can be addressed in parts relevant to cross-dataset validation. (4.7)

4.5 RD versus migrant phenology

The answer differs depending on the choice of metric to answer the question: migrant *peak* or *departure*. In the case of RD – Migrant peak DOY, there seems to be an equal split overall between positive and negative responses, but long distance migrants show a bias towards the latter. For the short-distance migrants, the non breeding sub populations are peaking before the RD, while they will affect the counts in the case of Lst. However, the migrant departure representation is yielding fairly constant negative results for RD – Migrant 50% threshold. No dataset or migratory strategy is being able to show that the RD ensures the survey effort is initiated after a major departure of the migrants. Although the metric’s definition is a bit sketchy, it was the most consistent signal recovered in the study: the RDs, as currently defined, are reached too early to reliably exclude the majority of the passage-migrant wave.

Table 5: Median (IQR) of the three metrics, in days, by dataset and migratory strategy. Negative = RD precedes the observed event and vice-versa. %neg = percentage of species in the distribution, yielding a negative response

Dataset	Zug	Ref – breeder peak		Ref – migrant peak		Ref – migrant 50%	
		Med. [IQR]	%neg	Med. [IQR]	%neg	Med. [IQR]	%neg
Bavaria	Kst	−25 [−34, 23]	64.5	15 [−40, 27]	42.1	−12 [−55, 11]	62.5
Bavaria	Lst	−21 [−36, −6]	93.3	−5 [−33, 4]	51.6	−15 [−38, −7]	88.0
Swiss migration	Kst	−39 [−71, −6]	77.8	1 [−33, 28]	50.0	−51 [−71, −17]	89.7
Swiss migration	Lst	−2 [−25, 4]	61.5	3 [−11, 8]	46.2	−27 [−32, −11]	85.7
BW I	Kst	11 [−19, 37]	40.0	3 [−40, 37]	48.9	−13 [−61, 9]	54.8
BW I	Lst	−11 [−48, −4]	76.5	−20 [−40, 6]	58.8	−14 [−23, −10]	87.5
BW II	Kst	5 [−38, 37]	46.0	5 [−38, 37]	47.6	−28 [−68, −3]	84.4
BW II	Lst	−14 [−25, −2]	77.3	−15 [−31, −3]	81.8	−31 [−44, −29]	90.9

4.6 Long term trends of migrant and breeder peaks

The 2007-2025 Swiss trend analysis (80 species: 43 Kst, 37 Lst) shows a consistent negative advancement for the breeder and migrant peak for the short distance migrants, while a diverged movement is seen for the long distance migrants, as shown in Table 6. The advancement is more pronounced for Kst in comparison to Lst. If we were to compare the same metric between the two migratory strategies, Kst have advanced their mating activity while Lst choose to delay by $0.15 \text{ days yr}^{-1}$, with 61.1% of positive responses. While Kst have advanced their breeder ($-0.55 \text{ days yr}^{-1}$) and migrant peak ($-0.57 \text{ days yr}^{-1}$) by similar magnitudes, long distance migrants have advanced their migration ($-0.19 \text{ days yr}^{-1}$), but have a considerable delay before they initiate breeding activity ($0.15 \text{ days yr}^{-1}$). Significant trends for individual species, although exceptions, are representative of the aggregated results for each metric×zug combination.

Table 6: Swiss 2007-2025 trend slopes of metrics (days yr^{-1}), median [IQR], by migratory strategy; %neg = advancing; %sig = statistical significance at $p < 0.05$.

Metric	N	Median [IQR]	%neg (advancing)	%sig
Breeder peak – Kst	43	−0.55 [−1.69, 0.55]	60.5	11.6
Breeder peak – Lst	36	0.15 [−0.13, 0.47]	38.9	5.6
Migrant peak – Kst	43	−0.57 [−1.77, 0.28]	67.4	23.3
Migrant peak – Lst	37	−0.19 [−0.44, 0.01]	70.3	5.4

4.7 Cross-dataset agreement (Q4)

Pearson correlation computed pairwise across the six possible dataset pairs separately by metric and migratory strategy. The mean r value has been taken from the heatmap and entered into Table 7. Similar to what was shown previously in 3, agreement is systemically stronger for the long distance migrants in comparison to the short distance migrants. The contrast is the most extreme for migrant peak DOY, where median r is 0.65 for Lst species but effectively zero (0.07) for Kst species, ie, the four datasets agree with each other on when Lst migrants peak, but not for the latter. These are from the results of the 16 species who are common to all the four datasets, but the pattern remains consistent across specific pair-wise comparisons. For other metrics as well, the amount of agreement for long-distance migrants have the edge and it is a bit close for breeder peak DOY and RD – breeder peak DOY.

Despite the generally weak results for Kst species, none were flagged as a consistent outlier (outside the 95% limits of agreement). Disagreement among datasets is diffused across the pool of short distance migrants and not driven by a handful of species. In reference to 4.1, the spread for both case-studies must be noted. For short-distance migrants, the cluster of the DOYs is in a narrower window as compared to Lst species, whose true peak dates are spread throughout a longer part of the season. Thus, a seemingly tight agreement for *Picus canus* is not defying the weak correlation seen in 7, but is a visualisation of an underlying effect.

Table 7: Median pairwise Pearson correlation (r) across the six pairwise dataset comparisons, by metric and migratory strategy

Metric	Kst	Lst
Migrant peak DOY	0.07	0.65
Breeder peak DOY	0.46	0.51
Ref – migrant peak	0.28	0.74
Ref – breeder peak	0.59	0.66

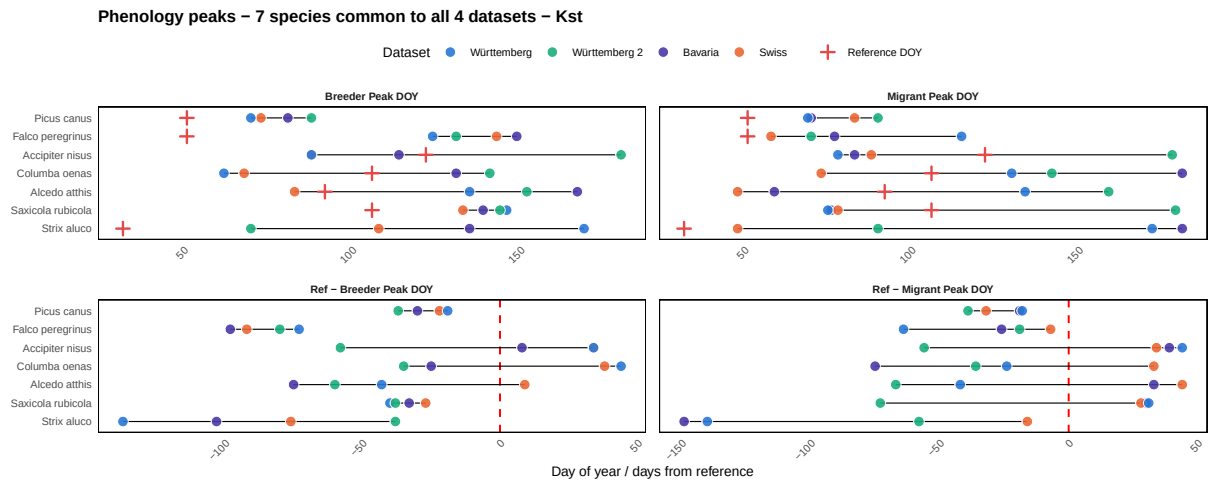
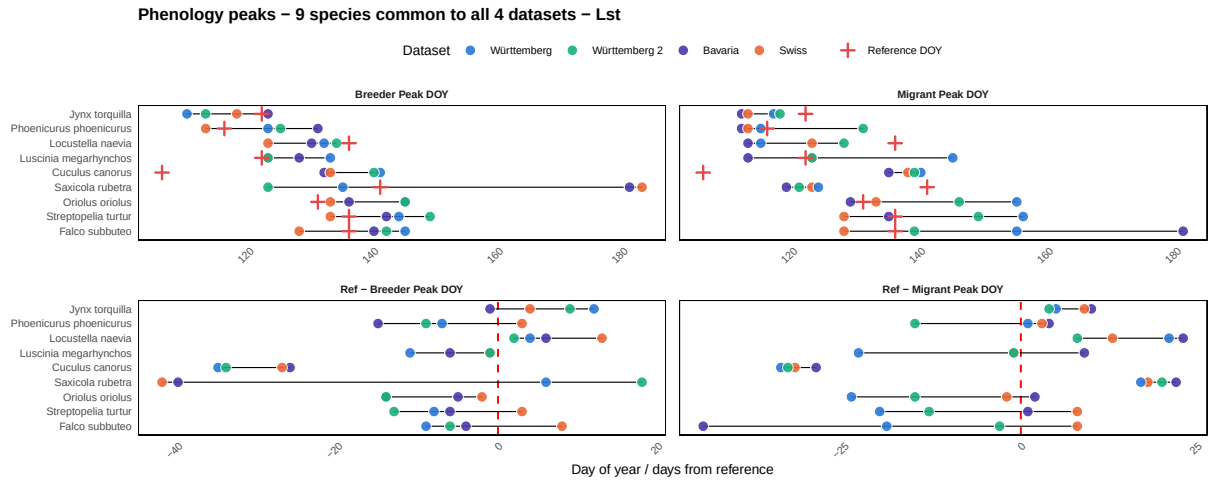


Figure 4: RD, breeder peak DOY, migrant peak DOY and the two derived metrics for the 16 species, which are common to all the four datasets, split by migratory strategy: short distance (Kst) and long distance migrants (Lst). Each row is one species, with the point colour denoting a specific dataset; red cross denoting RD and red dotted line denoting 0; horizontal line connecting the range of estimates across datasets for corresponding metric.

5 Conclusions

Q1 - Breeder phenology: Whether the RD precedes or follows peak breeding activity, depends on the migratory strategy. Long distance migrants show a robust pattern, cross-validated across datasets, wherein the breeder peak DOY was consistently arrived after the RD. However, the same doesn't apply to short distance migrants, where the results are genuinely mixed. Bavarian and Swiss datasets suggest that the RD is being able to capture the true peak, while Württemberg I and II datasets fail to do the same (a much closer split). The regional heterogeneity for Kst species can be an indicator on its own, suggesting any revision for their RDs might have to follow a spatial pattern, taking into consideration different habitats and elevational spread of the species. Table 5

Q2 - Migrant phenology: The RDs fail to align reasonably well with the timing of migrants' *departure* in almost every dataset and migratory strategy categories. While the peak of their activity is sometimes reached before the RD, taking RD - migrant 50% into consideration shows that the current RDs do not successfully lead to a survey which would be able to exclude non-breeders from the population. Surveys conducted shortly after the RDs are likely to include a substantial residual migrant component for most species, precisely the bias the RD concept was designed to avoid. This was the question where evidence for revision is the least ambiguous and most uniformly seen across datasets. Table 5

Q3 - Long term trends: Both breeder and migrant phenologies have advanced, on median over the 2007-2025 time period covered by the Swiss dataset for the short distance migrants. The assumption that long distance migrants, constrained by fixed departure cues, at distant wintering grounds are the ones lagging behind an advancing spring, while Kst species track local conditions more tightly and show a larger shift, is supported by the results in Table 6. However, these trend estimates are coming from a single dataset because of a sufficient temporal depth. Consequently, they should be treated as suggestive rather than conclusive generalisable results until a comparable multi-decade series is available from Bavaria or Württemberg.

Cross-dataset agreement: A clear result of this study is that the answers to Q1-Q3 are not uniformly reproducible across independently collected datasets. The degree of robustness itself varies systematically, depending on migratory strategy. Long distance migrants are much easier to classify consistently and show more consistent results across datasets, as seen in Table 7. Short distance migrants show weak median agreement for every metric, but this weak effect is diffused rather than concentrated. There are no consistent outliers being flagged, meaning the disagreement reflects genuinely noisier underlying signals rather than a handful of problematic species contaminating the dataset. Kst sub populations maybe smaller or have a greater sensitivity to the classification rules being used.

Compiling them together, these findings demonstrate that many reference dates warrant empirical reassessment, with respect to the migrant phenology (Q2). Moreover, any revision to the RDs should account for a degree of agreement (or disagreement) among independently monitored regional populations of the same species. For long-distance migrants, where cross-dataset validation is strong, a single revised RD per species is possible. For short-distance migrants, the combination of weak cross-dataset agreement and region-dependent results for Q1 suggests a need for locally calibrated RDs.

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Data and Code Availability

Analysis code and result outputs for all five component analyses are available at <https://github.com/hellodonki/Bird-Migration>. Raw monitoring data are held by the respective regional monitoring agencies and are available upon reasonable request subject to data-sharing agreements.